Milestone-2: Temporal Trend Analysis of US Disaster Declarations

Vinod Basavaraj Badiger

1. Project Description

The Visualizing US Natural Disaster Declarations project focuses on analysing historical disaster data to understand patterns and trends in federally declared disasters across the United States. The project uses the FEMA Disaster Declarations dataset, which contains detailed information about disaster events such as the type of disaster, affected states, declaration dates, and incident durations.

The objective of this milestone is to perform temporal trend analysis to study how disaster occurrences evolve over time. By analysing the dataset using data visualization techniques, the project aims to identify long-term trends, seasonal patterns, and differences in disaster types. Understanding these patterns can provide valuable insights into how disaster frequency and intensity change over time.

Through visual analysis of the dataset, the project highlights important insights such as increasing disaster occurrences, the dominance of certain disaster types, and seasonal disaster trends. These findings help improve understanding of natural disaster behaviour and support better disaster preparedness and policy planning.

2. Dataset Used

The dataset used in this project is the FEMA Disaster Declarations dataset, which contains historical records of disasters officially declared by the United States federal government. Each record in the dataset represents a disaster declaration event and provides information about the affected region, disaster category, and timeline of the incident.

The dataset includes several important attributes such as:

- State – The US state where the disaster occurred

- Incident Type – The category of disaster (e.g., flood, storm, wildfire, hurricane)
- Declaration Date – The date on which the disaster was officially declared
- Incident Begin Date – The start date of the disaster event
- Incident End Date – The end date of the disaster event
- Region – The FEMA region responsible for disaster management
- Fiscal Year Declared – The fiscal year in which the disaster was declared

The dataset contains tens of thousands of disaster records spanning multiple decades. This large dataset allows for meaningful trend analysis across time, enabling identification of long-term patterns and seasonal variations in disaster occurrences.

3. Environment Setup

The analysis for this project was conducted using Python in a Jupyter Notebook environment. Jupyter Notebook provides an interactive platform that allows code execution, visualization, and documentation in a single environment, making it ideal for data analysis and exploratory data science tasks.

Several Python libraries were used to perform data analysis and visualization:

- Pandas – Used for loading, cleaning, and manipulating the dataset. Pandas provides powerful data structures such as Data Frames that simplify data analysis tasks.

- NumPy – Used for efficient numerical computations and handling large datasets.

- Matplotlib – A fundamental plotting library used to create various types of visualizations such as line charts and bar charts.

- Seaborn – A high-level visualization library built on top of Matplotlib that provides aesthetically pleasing statistical plots.

These tools collectively provide a robust environment for performing data preprocessing, trend analysis, and visualization of disaster patterns.

4. Data Exploration

Before performing detailed analysis, the dataset was carefully explored to understand its structure and characteristics. Data exploration is an important step in any data analysis project because it helps identify potential data quality issues and ensures that the dataset is suitable for analysis.

The following steps were performed during the exploration phase:

1. Loading the dataset
   The dataset was loaded using the Pandas library to convert the raw CSV file into a structured Data Frame format.

2. Inspecting dataset dimensions
   The number of rows and columns in the dataset was examined to understand the dataset size.

3. Checking column data types
   The data types of each column were inspected to ensure that numeric, categorical, and date fields were correctly represented.

4. Identifying missing values
   Missing values in the dataset were detected using Pandas functions. This step helps determine whether data cleaning is required.

5. Viewing summary statistics
   Descriptive statistics were generated to understand the distribution of numerical features.

These steps ensured that the dataset was correctly structured and ready for further preprocessing and analysis.

5. Data Preprocessing

Before performing temporal analysis, the dataset required several preprocessing steps to ensure that time-related data was correctly formatted and usable for analysis.

One of the most important steps was converting the date columns into datetime format. In the raw dataset, date values were stored as text strings. These were converted into proper datetime objects so that time-based operations such as grouping by year or month could be performed.

The following preprocessing steps were implemented:

1. Conversion of date columns
   The declaration date, incident begin date, and incident end date were converted to datetime format using the Pandas to_datetime() function.

2. Extraction of temporal features
   From the declaration date column, two additional features were extracted:

   - Year – to analyze long-term disaster trends

- Month – to analyze seasonal patterns

3. Validation of date fields

   The dataset was checked for missing or invalid date values to ensure accurate analysis.

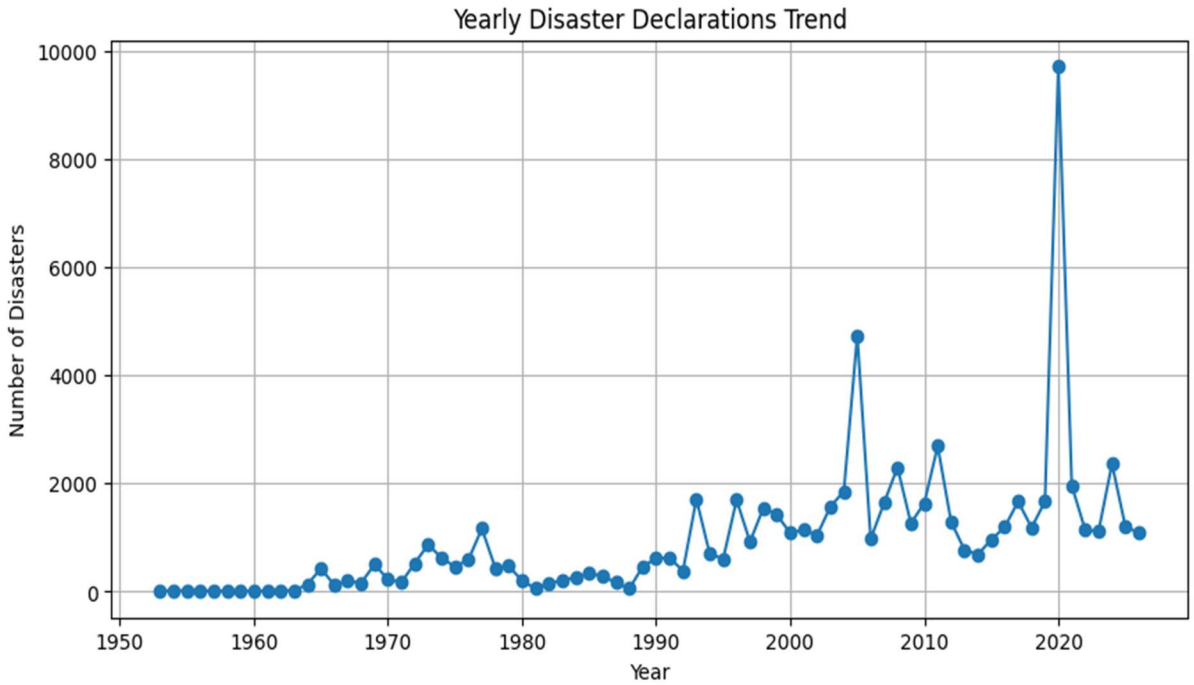
These preprocessing steps prepared the dataset for performing temporal grouping and visualization.

6. Yearly Disaster Trend Analysis

A time-series analysis was performed to understand how disaster declarations change over time.

The data was grouped by year and the number of disasters per year was calculated.

A line chart was created to visualize yearly disaster trends.



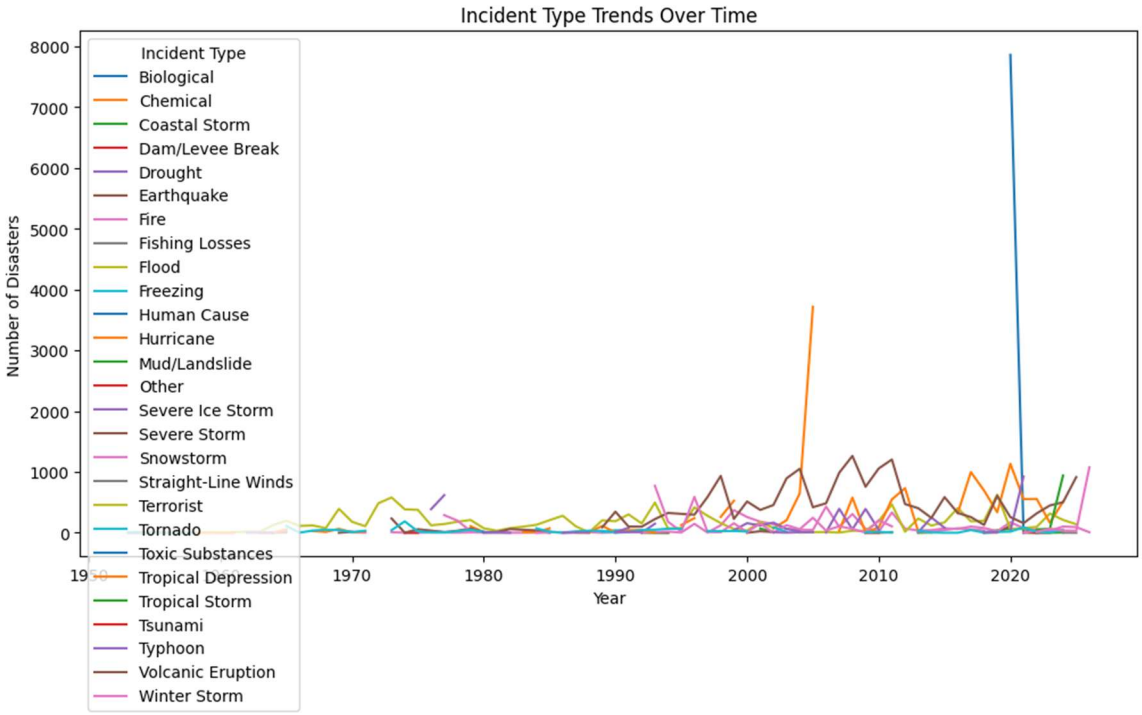
Observation:

The visualization helps identify increasing or decreasing patterns in disaster declarations over time.

7. Incident Type Trend Analysis

To compare disaster types across years, the dataset was grouped by both year and incidentType.

A multi-line chart was used to visualize trends for different disaster categories.



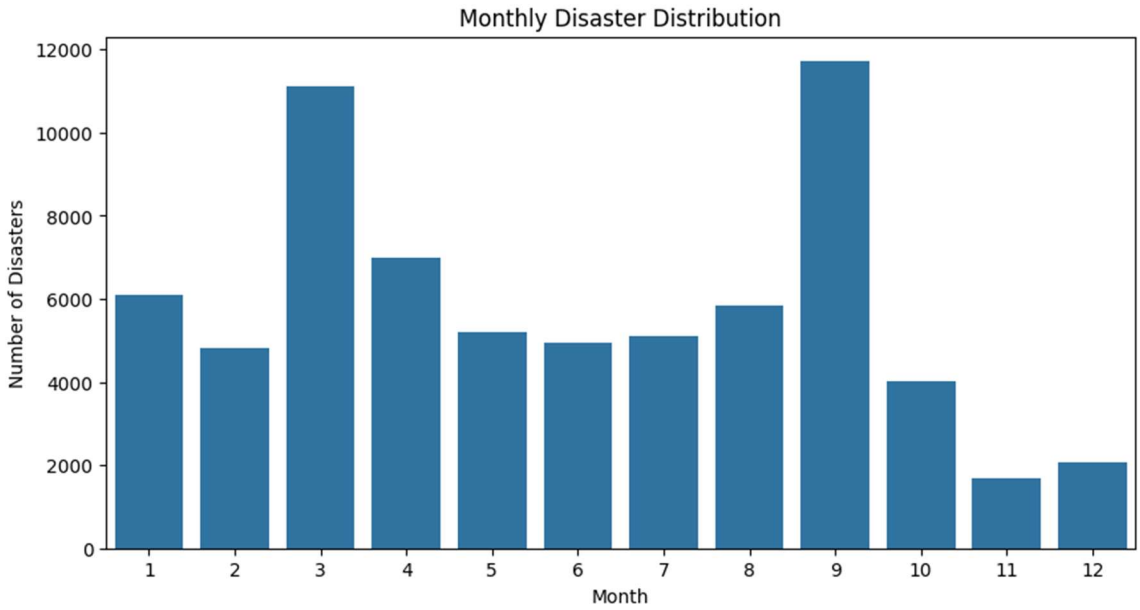
Observation:

Storms and floods appear frequently compared to other disaster types.

8. Monthly Seasonality Analysis

Seasonality analysis was performed by grouping disasters by month.

A bar chart was used to visualize the distribution of disasters across months.



Observation:

Certain months show higher disaster occurrences, indicating seasonal patterns.

9. Advanced Time Insights

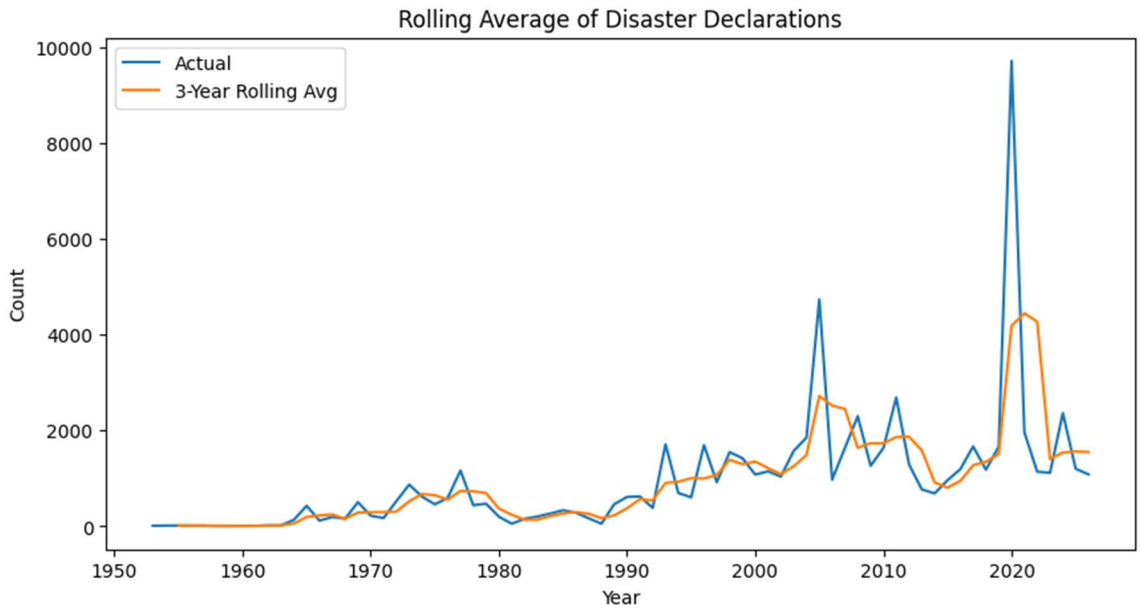
Additional analysis techniques were applied to extract deeper insights.

These include:

- Rolling average to smooth yearly fluctuations

- Heatmap to visualize disaster intensity across months and years

- Growth rate analysis for yearly changes



10. Conclusion

- The temporal analysis of the FEMA Disaster Declarations dataset provides valuable insights into the historical patterns of natural disasters in the United States.
- The analysis revealed that disaster declarations vary significantly across years and often show increasing trends in recent decades. Storms and floods appear to be among the most common disaster types, indicating the significant impact of weather-related events.
- Seasonal analysis also demonstrated that disasters tend to occur more frequently during certain months, suggesting strong links between disaster frequency and seasonal weather conditions.
- By combining data exploration, preprocessing, and visualization techniques, this milestone successfully identifies key temporal trends in disaster occurrences. These insights can support future disaster prediction models, policy development, and improved disaster preparedness strategies.